

PERSPECTIVE

Game changers: do new medications make lifestyle-based treatment of obesity obsolete?

Kristina H. Lewis^{1,2}  | Justin B. Moore^{1,2}  | Jamy D. Ard¹ 

¹Department of Epidemiology & Prevention, Division of Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, North Carolina, USA

²Department of Implementation Science, Division of Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, North Carolina, USA

Correspondence

Kristina H. Lewis, Department of Epidemiology & Prevention, Division of Public Health Sciences, Wake Forest University School of Medicine, 525 Vine St., Winston-Salem, NC 27101, USA.
Email: khlewis@wakehealth.edu

Abstract

Historically, obesity was viewed as a lifestyle disease, with an associated lifestyle solution, and approaches that embody the “eat less, move more” idea have dominated obesity treatment recommendations for over half a century. Meanwhile, the prevalence and severity of obesity continue to increase globally. Enter the so-called “game changers”: glucagon-like peptide-1 receptor agonists. In the media frenzy around these and other new antiobesity medications in the pipeline, lifestyle-based treatment researchers and practitioners may find themselves wondering whether behavioral approaches to obesity will become obsolete in this new therapeutic era. In this Perspective, the authors contend that medical approaches impact physiologic pathways to support the success of behavioral approaches. Similarly, behavioral approaches can improve weight loss-adjacent outcomes that are not addressed by medication. Thus, the two approaches are complementary and must coexist if we are to make a significant, population-level impact on the obesity epidemic.

BEHAVIORAL APPROACHES DO NOT PRODUCE LASTING WEIGHT LOSS FOR MOST PATIENTS

Decades of research have focused on modifying lifestyle behaviors to produce weight loss through reduced energy intake and increased exercise; however, lifestyle-based approaches to obesity treatment continue to demonstrate modest weight outcomes, with substantial heterogeneity in effectiveness and durability [1]. There are several important reasons why a purely behavioral approach to treating obesity is unlikely to produce clinically meaningful, sustained response for most patients.

First, lifestyle-based treatments do not address the physiology of obesity, i.e., the internal environment, encompassing the gut-brain connection and other systems involved in weight and appetite regulation. In the setting of energy restriction and weight loss, the body mounts counterregulatory physiologic cues that persist well

into the weight-maintenance phase [2–4]. Rather than identifying this physiology as the culprit, our field has often blamed a “lack of behavioral compliance” for poor treatment response or weight regain, contributing to stigma and shaming of those with the disease of obesity.

Second, individually focused behavioral interventions are unable to address the patient's external environment. Economic stability, access to resources, social and community context, and the built environment vary widely among the populations that we work with. Instead of addressing these drivers, lifestyle-focused interventions reach for more malleable targets such as nutritional knowledge, self-monitoring, and cooking skills. Unfortunately, for most patients, particularly those from historically marginalized groups or those living in lower-income areas, the obesogenic environment will overwhelm even substantial efforts that they make to enact individual behavior changes. Without equitable environment-level changes that address the social determinants of health, continued reliance on individual behavior changes to treat obesity may contribute to further increases in prevalence and widening disparities.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2023 The Authors. *Obesity* published by Wiley Periodicals LLC on behalf of The Obesity Society.

A related problem is the “research to practice” gap, in which even the most promising lifestyle interventions from randomized trials lose momentum when translated to clinical practice or community settings. This translational gap is unsurprising given a health care system whose payment models are misaligned for supporting lifestyle interventions, a medical workforce that lacks training on delivering lifestyle-based treatment, and weak public health infrastructure for community intervention delivery. Without major health system reform, as well as sustained investment to facilitate intervention delivery that approximates the fidelity seen in clinical trials, translating lifestyle interventions into real-world settings will continue to produce modest results [5,6].

ENTER THE “GAME CHANGERS”

Historically, the treatment of obesity has essentially meant a choice between lifestyle change on one end of the spectrum and bariatric surgery on the other. However, with newer antiobesity medications (AOMs) coming to market that boast impressive safety and weight-loss profiles, many patients and providers are starting to view medication as an attractive option to support long-term weight management, possibly without the need for much accompanying lifestyle intervention [7]. The SURMOUNT trial, for example, demonstrated greater than 20% mean reduction in weight at 72 weeks in a group of patients receiving a 15-mg weekly dose of tirzepatide, without a traditional intensive lifestyle intervention [7]. Although it has generally been assumed that combining more intensive behavioral intervention with AOMs will result in greater total weight loss (i.e., additive benefit), this may only be applicable to AOMs with lower average weight-loss effects [8]. When intensive lifestyle intervention was combined with semaglutide 2.4 mg in the STEP 3 trial [9], it did not produce a greater magnitude of weight loss than what was observed in the STEP 1 trial, at least in the short term.

Tirzepatide, semaglutide, and even older, non-glucagon-like peptide-1 receptor agonist (GLP-1RA) AOMs impact physiologic pathways and symptoms that otherwise make it difficult to initiate and sustain clinically significant weight loss. Additionally, although AOMs, like behavioral interventions, cannot change a patient's external environment, they may be able change one's response to that environment, resulting in durable weight loss despite the persistence of external challenges, much like bariatric surgery [10]. Given that the future of obesity treatment is likely to prominently feature AOMs, many are left wondering: what is the role for lifestyle and behavioral treatments moving forward?

WHERE DO WE GO FROM HERE? THE CASE FOR A NEW APPROACH

Even in this era of next generation AOMs, there is still a critical role for lifestyle and behavior change to improve population health outcomes. However, rather than being positioned as an obesity treatment strategy, lifestyle change interventions could be viewed as

a universal public health priority. There is strong evidence that behaviors including adoption of healthful dietary patterns, reducing sedentary time, engaging in regular physical activity, and receiving sufficient, good quality sleep have innumerable benefits, ranging from cardiovascular protection to improved mental health [11]. Therefore, a healthy lifestyle should be a goal that everyone, regardless of body mass index, should strive for and be given the resources to realistically achieve.

When we shift focus to the individual patient and consider how lifestyle change fits into the treatment paradigm for obesity, it is worth looking at how we manage other diet-related chronic diseases. For patients with type 2 diabetes or essential hypertension, physicians do not generally withhold medications while waiting for patients to “fail” dietary interventions, nor do they discontinue pharmacotherapy once a patient has reached a relevant clinical goal (e.g., antihypertensives are not stopped when a patient reaches a blood pressure of 120/80). Rather, treatment intensity is matched with the severity of the disease at the time of presentation, and this often means starting pharmacotherapy at the outset [12,13]. Similarly, effective medical or surgical treatment of obesity should not be withheld pending a patient's behavioral “failure” or discontinued in the setting of reaching a weight goal because the underlying disease of obesity (much like hypertension) has not been “cured” *per se*. For patients who are not interested in weight loss, lifestyle intervention may be a sufficient intervention for preventing additional weight gain. Weight maintenance is a salient and clinically meaningful outcome that has shown promise in diverse populations at high risk of health disparities [14].

Finally, lifestyle behavioral therapy could help us achieve “quality” in weight loss, not just the “quantity” that is so appealing with modern AOMs. Behavioral therapy has previously been viewed as a way to keep patients in a negative energy balance. With that process substantially empowered by AOMs, lifestyle behavioral therapy could be refocused as a tool to optimize nutrition and physical activity in an attempt to improve overall health during weight loss. For example, higher protein intake with lifestyle-based treatment is associated with improved body composition during active weight loss and could be applicable during AOM use to optimize loss of fat mass with preservation of lean mass [15], or, while a patient is comfortably losing weight with an AOM, this could free up time to focus energy on improving cardiovascular fitness with increased levels of exercise.

Ultimately, AOMs will make lower body weights more attainable and sustainable, but they will not address the widespread pathologies of our food and lived environments. Healthy lifestyles for all people, irrespective of body mass index, should be supported using individual, policy, systems, and environmental approaches that address the social determinants of health and are responsive to community needs. Emerging evidence has suggested that some of these approaches (e.g., taxation of sugary drinks, school environment changes, food subsidies or programs to provide healthful foods) may be effective in impacting behaviors and weight outcomes [16–22]. Policy changes must not neglect the medical system, in which reimbursement metrics and the training of medical professionals could be modified to better incentivize well care and healthy lifestyle promotion for all patients, not just those with obesity.

Similarly, policy changes around the pricing of AOMs must be considered to ensure equitable uptake of these tools. Finally, as a field, obesity medicine must move away from the toxic “lifestyle versus medical therapy” debate. Our patients will benefit most if we can learn to pair lifestyle interventions with pharmacotherapy to both optimize health outcomes and help them maintain lower body weights. 

ACKNOWLEDGMENTS

The authors would like to thank Drs. Mara Vitols and Joseph Skelton for providing valuable feedback on an earlier draft of this manuscript.

CONFLICT OF INTEREST STATEMENT

Jamy D. Ard reports personal fees and nonfinancial support from Nestlé Healthcare Nutrition; grants and personal fees from Eli Lilly and Company; grants from Epitome Medical Ltd.; grants and personal fees from UnitedHealth Group Inc.; personal fees from Novo Nordisk A/S; nonfinancial support from KVK Tech; personal fees and nonfinancial support from WW (formerly WeightWatchers); personal fees from Regeneron Pharmaceuticals, Inc.; personal fees from Brightseed Bioactives; grants and personal fees from Boehringer Ingelheim; and personal fees from Intuitive Biosciences outside the submitted work. Kristina H. Lewis reports nonfinancial support from KVK Tech outside the submitted work. Justin B. Moore reports that he provides consulting services for Medtronic and serves on the scientific advisory board of Heali AI.

ORCID

Kristina H. Lewis  <https://orcid.org/0000-0002-4921-7230>

Justin B. Moore  <https://orcid.org/0000-0003-4059-0538>

Jamy D. Ard  <https://orcid.org/0000-0002-1643-6795>

REFERENCES

1. LeBlanc ESPC, Webber EM, Redmond N, Rushkin M, O'Connor EA. Behavioral and pharmacotherapy weight loss interventions to prevent obesity-related morbidity and mortality in adults: updated evidence report and systematic review for the US Preventive Services Task Force. *JAMA*. 2018;320(11):1172-1191.
2. MacLean PS, Higgins JA, Giles ED, Sherk VD, Jackman MR. The role for adipose tissue in weight regain after weight loss. *Obes Rev*. 2015; 16(Suppl 1):45-54.
3. Apovian CM, Aronne LJ, Bessesen DH, et al. Pharmacological management of obesity: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2015;100(2):342-362.
4. Sumithran P, Prendergast LA, Delbridge E, et al. Long-term persistence of hormonal adaptations to weight loss. *N Engl J Med*. 2011; 365(17):1597-1604.
5. Ard J. Obesity in the US: what is the best role for primary care? *BMJ*. 2015;350:g7846.
6. de Lannoy L, Cowan T, Fernandez A, Ross R. Physical activity, diet, and weight loss in patients recruited from primary care settings: an

- update on obesity management interventions. *Obes Sci Pract*. 2021; 7(5):619-628.
7. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387(3):205-216.
8. Wadden TA, Foreyt JP, Foster GD, et al. Weight loss with naltrexone SR/bupropion SR combination therapy as an adjunct to behavior modification: the COR-BMOD trial. *Obesity (Silver Spring)*. 2011; 19(1):110-120. doi:10.1038/oby.2010.147
9. Wadden TA, Bailey TS, Billings LK, et al. Effect of subcutaneous semaglutide vs placebo as an adjunct to intensive behavioral therapy on body weight in adults with overweight or obesity: the STEP 3 randomized clinical trial. *JAMA*. 2021;325(14):1403-1413.
10. Drewnowski A, Hong BD, Shen E, et al. Neighborhood deprivation and residential property values do not affect weight loss at 1 or 3 years after bariatric surgery. *Obesity (Silver Spring)*. 2023;31(2):545-552.
11. Dietary Guidelines Advisory Committee. *Scientific Report of the 2020 Dietary Guidelines Advisory Committee: Advisory Report to the Secretary of Agriculture and the Secretary of Health and Human Services*. 2020. US Government.
12. American Diabetes Association. Standards of Care in Diabetes-2023 abridged for primary care providers. *Clin Diabetes*. 2022;41(1):4-31.
13. Cifu AS, Davis AM. Prevention, detection, evaluation, and management of high blood pressure in adults. *JAMA*. 2017;318(21):2132-2134.
14. Bennett GG, Foley P, Levine E, et al. Behavioral treatment for weight gain prevention among black women in primary care practice: a randomized clinical trial. *JAMA Intern Med*. 2013;173(19):1770-1777.
15. McCarthy D, Berg A. Weight loss strategies and the risk of skeletal muscle mass loss. *Nutrients*. 2021;13(7):2473.
16. Micha R, Karageorgou D, Bakogianni I, et al. Effectiveness of school food environment policies on children's dietary behaviors: a systematic review and meta-analysis. *PloS One*. 2018;13(3):e0194555.
17. Krieger J, Bleich SN, Scarmo S, Ng SW. Sugar-sweetened beverage reduction policies: Progress and promise. *Annu Rev Public Health*. 2021;42:439-461.
18. Niebylski ML, Redburn KA, Duhaney T, Campbell NR. Healthy food subsidies and unhealthy food taxation: a systematic review of the evidence. *Nutrition*. 2015;31(6):787-795.
19. Lipek T, Igel U, Gausche R, Kiess W, Grande G. Obesogenic environments: environmental approaches to obesity prevention. *J Pediatr Endocrinol Metab*. 2015;28(5-6):485-495.
20. Kenney EL, Barrett JL, Bleich SN, Ward ZJ, Cradock AL, Gortmaker SL. Impact of the healthy, hunger-free kids act on obesity trends. *Health Aff*. 2020;39(7):1122-1129.
21. Daepf MIG, Gortmaker SL, Wang YC, Long MW, Kenney EL. WIC food package changes: trends in childhood obesity prevalence. *Pediatrics*. 2019;143(5):e20182841.
22. Patel AI, Schmidt LA, McCulloch CE, et al. Effectiveness of a school drinking water promotion and access program for overweight prevention. *Pediatrics*. 2023;152(3):e2022060021.

How to cite this article: Lewis KH, Moore JB, Ard JD. Game changers: do new medications make lifestyle-based treatment of obesity obsolete? *Obesity (Silver Spring)*. 2024;32(2): 237-239. doi:10.1002/oby.23962